

Kushal Erramilli

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🌐 [kushalerramilli](https://kushalerramilli.github.io)

Education

University of Maryland, Baltimore County (UMBC)

Master of Science in Data Science | GPA: 3.9/4.0

Expected Dec 2026

Baltimore, MD

Malaviya National Institute of Technology (MNIT)

Bachelor of Technology in Computer Science and Engineering

May 2023

Jaipur, India

Experience

Carrier Corporation

July 2023 – Dec 2024

Management Trainee, Engineering Leadership Program

- Implemented C++ event-handling routines for threshold alarms and mode-trigger logic in embedded firmware. Authored multiple functions, tested across hardware setups, and delivered code through peer-reviewed releases.
- Developed a Python tool for Wi-Fi authentication and real-time monitoring of remote embedded controllers, automating connectivity diagnostics.
- Analyzed industrial sensor datasets using Python and Pandas. Generated Jupyter-based reports covering fault-detection modules, edge-case validation, and findings for clients.
- Built reusable Python libraries for standardized engineering calculations, allowing teams to apply shared frameworks with project-specific configurations.

Projects

RAG Document Intelligence Engine | Python, FastAPI, Next.js, ChromaDB, OpenAI, Groq, Docker, AWS EC2/ECR | GitHub

- Built production-grade RAG pipeline with intelligent query routing: simple queries use direct vector search (~400ms), while complex queries use HyDE + cross-encoder reranking (~1300ms); implemented real-time token streaming via SSE with per-claim source citations and page numbers.
- Benchmarked system performance using the RAGAS evaluation framework across 16 parameter sweep experiments, achieving a 0.90 average score in faithfulness, relevancy, and context recall; deployed using Docker multi-stage builds on AWS EC2 + ECR with per-user API key architecture and zero server-side credential storage.

CareerCoach – AI-Powered Career Guidance Platform | Next.js, Tailwind CSS, Node.js, Prisma, PostgreSQL, Google AI

- Designed an intuitive responsive UI for a full-stack platform that combines Google AI to provide industry information, suited career paths, and developed skills, along with AI interviewers and customized feedback systems to prepare users for tailored performances.
- Engineered AI-powered resume and cover letter generators, streamlining applications by producing tailored documents based on user profiles and job descriptions.

Leaf Disease Detection using Attention-Based CNNs | Python, PyTorch, Deep Learning

- Designed a QuadNET attention module yielding 8–10% increased classification accuracy over baseline CNNs on a 50K+ leaf image dataset and benchmarked against attention mechanisms (CBAM, SE-Net), achieving superior F1-scores and robustness.
- Provided insights for enabling early detection of crop diseases and decreasing manual crop inspection costs for agri-tech use cases.

Certifications

AWS Certified Cloud Practitioner

Mar 2026 – Mar 2029

Amazon Web Services (AWS)

- Credential ID: 2ed4e3144bdc4518bccffd41f21dc9bc | Verify Credential

Technical Skills

Programming: Python, C++, JavaScript, SQL

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Systems

Backend & Web: Node.js, Express.js, Next.js, REST APIs

Systems & Tools: Git, Linux, Jupyter

Databases: MySQL

Cloud: AWS (EC2, S3, RDS)